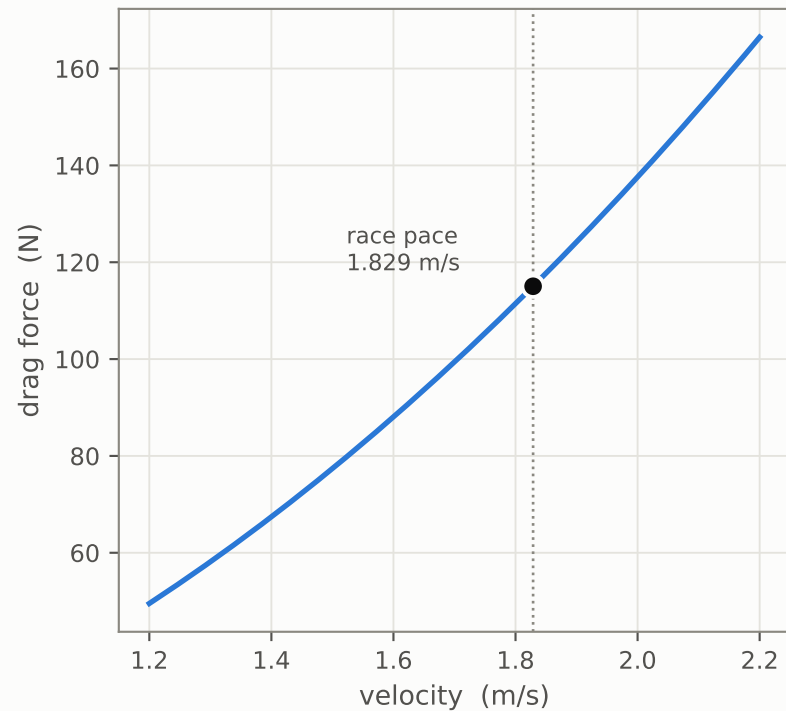
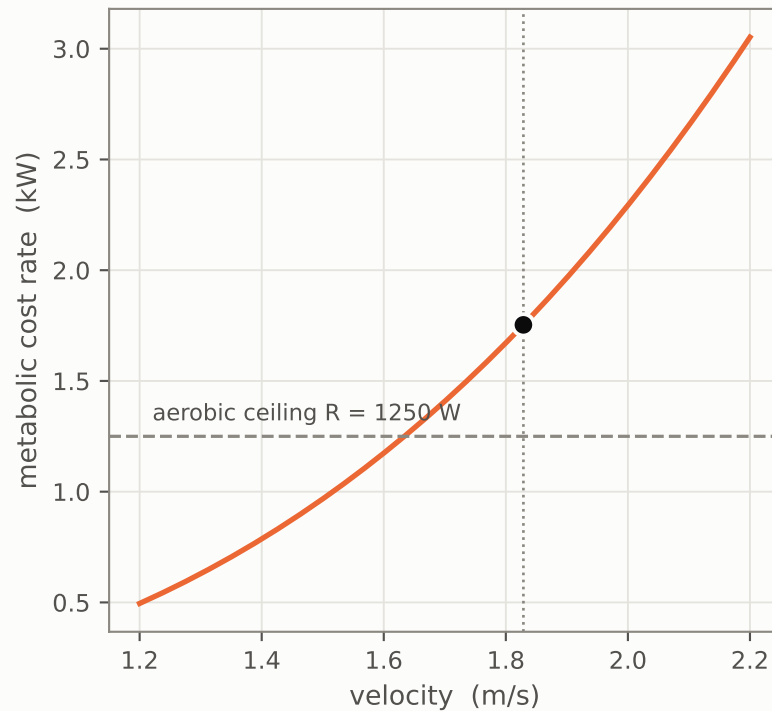


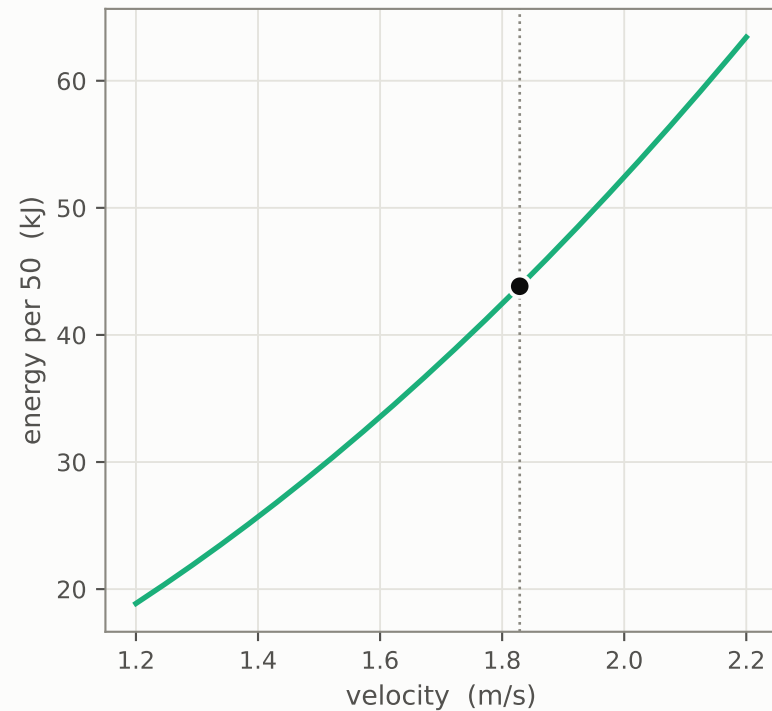
$$F_D = \frac{1}{2}\rho C_D A v^2$$



$$C(v) = kv^3$$



$$C(v) \cdot t = kdv^2 \quad (\text{cost of one split})$$



reference_M1518: $\frac{1}{2}\rho C_D A = 34.4 \text{ N s}^2/\text{m}^2$, $k = 286.6 \text{ W s}^3/\text{m}^3$, $p = 3.0$, $\eta = \eta_p \cdot \eta_g = 0.120$.